The objective of this project was to identify the optimal neural network architecture for classifying indoor scene styles. We benchmarked four distinct architectures representing different paradigms in computer vision:

1. **ResNet:** Standard Convolutional Neural Networks (CNN).
2. **ConvNeXt:** Modernized, Transformer-inspired CNN.
3. **ViT (Vision Transformer):** Pure attention-based architecture.
4. **DINOv3:** Self-supervised Foundation Model.

Key Finding: Contrary to initial hypotheses regarding efficient CNNs or Self-Supervised Learning, the **Vision Transformer (ViT-Large)** demonstrated superior performance, achieving the highest Private Score of **0.39985**. While traditional CNNs (ResNet) provided a strong baseline, and modernized CNNs (ConvNeXt) offered competitive results, the pure attention mechanism of the ViT proved most capable of capturing the complex, global semantic relationships required for style classification.

Architecture Breakdown

A. ResNet (Residual Networks)

- **Role:** The Baseline.
- **Variants Tested:** ResNet-50.
- **Training Duration:** 8 Epochs (Full Dataset).
- **Rationale:** Utilized as a standard benchmark to determine if modern architectures offer a tangible upgrade over established, robust heuristics.

B. ConvNeXt

- **Role:** The Modern CNN.
- **Variants Tested:** ConvNeXt-Tiny (50 Epochs), ConvNeXt-Base (10 Epochs).
  - **Rationale:** To test if the "Sliding Window" inductive bias of CNNs, when updated with Transformer-like design principles (large kernels, LayerNorm), could outperform pure Transformers in texture recognition.

C. Vision Transformer (ViT)

- **Role:** The Pure Transformer.
- **Variants Tested:** ViT-Base (5 Epochs), ViT-Large (15 Epochs).
- **Rationale:** To evaluate the effectiveness of Global Self-Attention. Style is a global property (the "vibe" of a room), utilizing ViT's ability to relate distant pixels (e.g., matching a ceiling light to a floor rug) without being constrained by the local receptive fields of CNNs.

D. DINOv3

- **Role:** The Self-Supervised Foundation Model.
- **Variants Tested:** DINOv3 (8 Epochs).
- **Rationale:** To test if a model pre-trained on billions of images via Self-Supervised Learning (SSL) could adapt to style classification with minimal fine-tuning, leveraging its pre-existing understanding of geometry.

Trials and Results

Trial 1: ResNet-50 (Baseline)

- **Experiment:** Full dataset training for 8 epochs.
- **Results:**
  - **Private Score:** 0.36724
  - **Public Score:** 0.35964
- **Analysis:** ResNet-50 performed surprisingly well, outperforming the smaller ConvNeXt-Tiny and competing closely with ConvNeXt-Base. This suggests that for this specific dataset size, the inductive bias of standard CNNs remains a very robust and data-efficient solution.

Trial 2: ConvNeXt

- **Experiment:**
  - *Tiny:* 50 Epochs (Train/Val split).
  - *Base:* 10 Epochs (Full Dataset).
- **Results:**
  - **ConvNeXt-Tiny:** Private 0.34540 / Public 0.32010
  - **ConvNeXt-Base:** Private 0.36544 / Public 0.34725
- **Analysis:** ConvNeXt-Base showed strong performance but failed to decisively beat the older ResNet-50 baseline. The Tiny variant, despite a massive training schedule (50 epochs), struggled to generalize, likely due to limited model capacity relative to the complexity of the style features. The "modernized" CNN features did not provide a significant edge over standard residual connections in this context.

Trial 3: Vision Transformer (ViT)

- **Experiment:**
  - *Base:* 5 Epochs.
  - *Large:* 15 Epochs.
- **Results:**
  - **ViT-Base:** Private 0.33210 / Public 0.32948
  - **ViT-Large: Private 0.39985 / Public 0.36727 (Winner)**
- **Analysis:** ViT-Large was the clear winner. The jump in accuracy from Base to Large indicates that "Style" is a high-complexity problem requiring massive capacity. The 15-epoch training allowed the Global Attention mechanism to fully stabilize. Unlike CNNs, which focus on

local textures, the ViT likely succeeded by modeling the *composition* of the room—a crucial factor in distinguishing styles like "Boho" from "Eclectic."

Trial 4: DINOv3

- **Experiment:** Fine-tuning for 8 Epochs.
- **Results:**
  - **Private Score:** 0.23898
  - **Public Score:** 0.22863
- **Analysis:** This was an experimental disappointment. Despite its foundation model status, DINOv3 failed to adapt to the specific nuances of interior design styles. The domain shift between its pre-training objective (object/geometry understanding) and our objective (abstract style recognition) may have been too large to bridge in just 8 epochs, or the self-supervised features were not discriminative enough for these specific classes compared to the supervised pre-training of the ViT/ResNet models.

Performance Comparison

Model	Configuration	Private Score	Public Score
ViT-Large	**15 Epochs**	**0.39985**	**0.36727**
ResNet-50	8 Epochs	0.36724	0.35964
ConvNeXt-Base	10 Epochs	0.36544	0.34725

Model	Configuration	Private Score	Public Score
ConvNeXt-Tiny	50 Epochs	0.34540	0.32010
ViT-Base	5 Epochs	0.33210	0.32948
DINOv3	8 Epochs	0.23898	0.22863

Conclusion & Explanations

Based on these trials, we draw the following conclusions:

1. **Capacity is King:** The significant gap between ViT-Large (0.399) and all other models (hovering around 0.32–0.36) suggests that the problem requires a high-capacity model to capture subtle stylistic differences. Smaller models (ConvNeXt-Tiny, ViT-Base) underfit the complexity of the task.

2. **Global vs. Local:** The success of ViT-Large over strong CNNs (ResNet/ConvNeXt) confirms that **Attention-based mechanisms are better suited for Style Classification.** Style is defined by the relationship between objects (e.g., a rustic table *plus* an industrial lamp), which global attention captures better than local convolutions.

3. **Supervised vs. Self-Supervised:** For this specific task, standard supervised pre-training (ResNet/ViT) vastly outperformed self-supervised learning (DINOv3). The "vibe" of a style is a human-labeled concept that likely requires supervised alignment rather than purely geometric understanding.

Final Recommendation

ViT-Large is selected as the primary architecture. Despite the higher computational cost of training (15 epochs on a large model), it provided the only statistically significant leap in performance, breaking the ~0.36 ceiling hit by all CNN-based approaches. Future work should focus on regularizing ViT-Large to further improve the Public/Private score consistency.